



## **MEDTRACK DV**

# **Hospital Operations and Patient Analytics Dashboard**

### **Project Documentation**

Developed Using  
Tableau | Excel/CSV | Anaconda

### **Submitted By**

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## Project Overview

**MedTrack DV – Hospital Operations and Patient Analytics Dashboard** is a healthcare analytics and data visualization project designed to analyze hospital operations, patient activity, doctor performance, and revenue using interactive dashboards.

The project transforms hospital operational data into meaningful visual insights using Tableau. The dashboard provides an interactive environment where users can monitor important healthcare performance indicators, compare operational trends, analyze patient characteristics, evaluate doctor-related metrics, and understand revenue distribution.

The project is organized into four major dashboard views: Overview, Doctor Analysis, Patient Analysis, and Revenue Analysis. The Overview dashboard provides a high-level summary of hospital performance through key performance indicators and visualizations. The Doctor Analysis dashboard focuses on doctor workload, doctor types, status, and case-related metrics. The Patient Analysis dashboard provides insights into patient demographics, case types, payer distribution, nationality, length of stay, and case complexity. The Revenue Analysis dashboard focuses on revenue trends and revenue distribution across specialties, case types, payers, and other relevant dimensions.

The dashboard uses interactive filters and navigation features to allow users to explore the data from different perspectives. This helps convert raw hospital records into a structured analytical solution that can support operational monitoring and data-driven decision-making.

### Project Workflow

Raw Hospital Data → Data Preparation → KPI Definition → Data Analysis → Tableau Visualizations → Interactive Dashboards → Insights → Deployment

| Main Dashboard Areas |                                               |
|----------------------|-----------------------------------------------|
| Dashboard            | Purpose                                       |
| Overview             | Overall hospital performance                  |
| Doctor Analysis      | Doctor-related workload and performance       |
| Patient Analysis     | Patient demographics and operational patterns |
| Revenue Analysis     | Financial and revenue-related analysis        |

## Project Objectives

The main objective of the **MedTrack DV – Hospital Operations and Patient Analytics** is to transform hospital operational data into meaningful and interactive visual insights using Tableau. The project focuses on improving the understanding of hospital performance, patient activity, doctor-related metrics, and revenue patterns.

### Key Objectives

1. **To analyse hospital operational data** and identify important patterns related to admissions, patient activity, length of stay, severity, and case complexity.
2. **To develop interactive dashboards** that provide a clear and user-friendly view of hospital performance.
3. **To monitor key performance indicators (KPIs)** such as total cases, total revenue, average Length of Stay (LOS), average CMI, and severity-related metrics.
4. **To analyse doctor-related performance** by examining doctor workload, doctor type, doctor status, case distribution, and case complexity.
5. **To understand patient characteristics and patterns** through analysis of gender, nationality, case type, payer/insurance, specialty, and other patient-related attributes.
6. **To analyse hospital revenue** across specialties, case types, payers, gender, and monthly trends to understand the major contributors to revenue.
7. **To provide interactive filtering and navigation** so users can explore specific periods, specialties, genders, case types, doctors, and other relevant categories.
8. **To present complex healthcare data in an easy-to-understand visual format**, enabling hospital stakeholders to identify trends, compare performance, and support data-driven decision-making.
9. **To create a portfolio-ready dashboard solution** with proper documentation, organized project structure, and deployment through platforms such as GitHub and Tableau Public.

## Dataset Description & Sources

The MedTrack DV – Hospital Performance Dashboard uses a hospital operations dataset obtained from Kaggle, an online platform for datasets and data science projects. The dataset contains information related to patient cases, doctors, medical specialties, insurance/payers, case types, clinical characteristics, discharge details, and revenue. It is used to analyze hospital operations and identify meaningful patterns in patient activity, doctor performance, case complexity, length of stay, and financial performance.

Each record in the dataset represents a hospital case/admission and contains multiple attributes associated with that case. The major fields used in the project include Case\_No, Gender, Nationality, DoctorName, DoctorLicense, Doctor Type, Doctor Status, Specialty, CMI Value, Insurance/Payer, Case Type, LOS, Severity, Discharge Time, Discharge Before 12PM, and Revenue.

The dataset fields are mainly categorized as follows:

- Patient Information: Gender, Nationality, DOB
- Doctor Information: DoctorName, DoctorLicense, Doctor Type, Doctor Status
- Clinical Information: Specialty, CMI Value, Severity, LOS, Surgical Mix
- Case Information: Case\_No and Case Type
- Insurance Information: Insurance/Payer
- Financial Information: Revenue
- Discharge Information: Discharge Time and Discharge Before 12PM
- Time Information: Month

### Data Source

The original dataset was downloaded from Kaggle and was available in a structured data format suitable for analysis. The dataset was reviewed and prepared before being used for dashboard development. Data preparation included checking data types, handling missing or inconsistent values, and organizing the required fields for analysis. The cleaned dataset was then connected to Tableau for creating interactive visualizations and dashboards.

The dataset is used for data analysis and visualization purposes. It is not intended to represent confidential or identifiable real-world patient records, and the dashboard is designed to demonstrate healthcare operational analysis.

## Data Preparation

The hospital dataset downloaded from **Kaggle** was prepared before creating the Tableau dashboards. Data preparation was performed to improve data quality, maintain consistency, and ensure that the fields could be correctly used for analysis and visualization.

The preparation process included the following steps:

1. **Data Collection:** The hospital dataset was downloaded from Kaggle and imported into a structured format for analysis.
2. **Data Inspection:** The dataset was examined to understand its columns, data types, missing values, duplicate records, and overall structure.
3. **Data Cleaning:** Missing, inconsistent, or incorrectly formatted values were identified and handled wherever required. Unnecessary fields and values that were not relevant to the dashboard analysis were avoided.
4. **Data Type Validation:** Fields such as Case\_No, Revenue, LOS, CMI Value, and Severity were checked to ensure that numerical values were recognized correctly. Date and categorical fields were also reviewed for appropriate data types.
5. **Missing Value Handling:** Missing values in important fields such as insurance information and discharge-related fields were reviewed and treated appropriately so that they would not cause errors during visualization.
6. **Consistency Checking:** Category values such as Gender, Specialty, Doctor Type, Doctor Status, Case Type, and Insurance/Payer were checked for consistency to avoid duplicate categories caused by different spellings or formats.
7. **Data Validation:** The cleaned data was checked to ensure that important measures such as case count, revenue, average LOS, and CMI values produced reasonable results.
8. **Tableau Integration:** After preparation, the cleaned dataset was connected to Tableau, where calculated measures, KPIs, charts, filters, and interactive dashboards were developed.

The prepared dataset provided a reliable foundation for developing the Overview, Doctor Analysis, Patient Analysis, and Revenue Analysis dashboards.

## Healthcare Operations Methodology

The Healthcare Operations Methodology defines the approach used to transform hospital operational data into meaningful insights through the MedTrack DV dashboard. The methodology focuses on analyzing patient activity, doctor operations, clinical complexity, hospital utilization, and revenue performance using Tableau.

The process begins with collecting the hospital dataset from Kaggle, followed by data inspection, cleaning, validation, and preparation. Relevant healthcare measures and dimensions are then selected based on the requirements of hospital operational analysis.

The prepared data is analyzed using important measures such as Case Count, Revenue, Length of Stay (LOS), CMI Value, and Severity. Case Count is used to understand hospital workload and admission volume. LOS is analyzed to understand the duration of patient stays and hospital resource utilization. CMI Value is used as an indicator of the relative complexity and resource intensity of cases. Revenue is analyzed to understand financial performance across different hospital categories.

The methodology is divided into four major analytical areas:

- **Overview Analysis:** Provides a high-level view of hospital performance using KPIs and charts for cases, revenue, LOS, CMI, severity, specialty, gender, and case type.
- **Doctor Analysis:** Examines doctor-related metrics such as case workload, doctor type, doctor status, CMI, revenue, and specialty distribution.
- **Patient Analysis:** Analyzes patient-related characteristics including gender, nationality, case type, payer/insurance, LOS, specialty, and case complexity.
- **Revenue Analysis:** Evaluates revenue trends and compares revenue across specialties, case types, payers, gender, and other relevant dimensions.

Tableau visualizations such as bar charts, line charts, pie/donut charts, treemaps, maps, and scatter plots are selected according to the type of analysis required. Interactive filters for month, specialty, gender, case type, doctor, and other dimensions allow users to explore specific portions of the dataset.

## KPI Definitions

Key Performance Indicators (KPIs) are numerical measures used in the **MedTrack DV – Hospital Operations and Patient Analytics** to provide a quick overview of hospital operations and performance. The selected KPIs represent important aspects such as case volume, financial performance, patient stay, case complexity, and severity.

| KPI                      | Definition                                              | Purpose                                                                                                   |
|--------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Total Cases</b>       | Count of hospital case/admission records using Case_No. | Measures overall hospital case volume and operational workload.                                           |
| <b>Total Revenue</b>     | Sum of the Revenue values for all cases.                | Measures the overall revenue generated by the hospital cases.                                             |
| <b>Average LOS</b>       | Average of the LOS (Length of Stay) values.             | Indicates the average duration patients remain in the hospital and helps understand resource utilization. |
| <b>Average CMI Value</b> | Average of the CMI Value across cases.                  | Indicates the average relative complexity and resource intensity of the cases handled.                    |
| <b>Total Severity</b>    | Sum of the numerical Severity values.                   | Provides an overall measure of the severity values recorded across hospital cases.                        |

### Doctor Analysis KPIs

| KPI                         | Definition                                                 | Purpose                                                                |
|-----------------------------|------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Doctor Count</b>         | Count of doctor license records.                           | Provides an overview of the doctor records represented in the dataset. |
| <b>Doctor-related Cases</b> | Count of Case_No associated with doctors.                  | Measures the case workload represented for doctors.                    |
| <b>Average CMI</b>          | Average CMI value for the selected doctor-related records. | Helps compare case complexity across doctor categories.                |

## **Patient & Revenue Analysis Metrics**

The Patient Analysis dashboard uses operational and clinical metrics to understand patient characteristics, hospital utilization, and case complexity. Average Length of Stay (LOS) measures the average number of days patients stay in the hospital and helps evaluate bed utilization and operational efficiency. Average CMI Value (Case Mix Index) indicates the average relative complexity and resource intensity of the cases handled by the hospital. Patient-related metrics are further analyzed across dimensions such as gender, nationality, specialty, case type, and insurance/payer to identify differences in patient distribution and healthcare utilization.

The Revenue Analysis dashboard focuses on understanding the financial performance of hospital cases. Total Revenue represents the combined revenue generated from all hospital cases in the selected dataset or filter context. Average Revenue represents the average revenue associated with the hospital cases and helps compare the typical financial value of cases. Revenue is analyzed across medical specialties, case types, insurance/payer categories, gender, and time periods to identify the major contributors to hospital revenue. The Monthly Revenue Trend helps monitor changes in revenue over time, while comparisons across specialties and case types help identify areas with higher or lower revenue contribution.

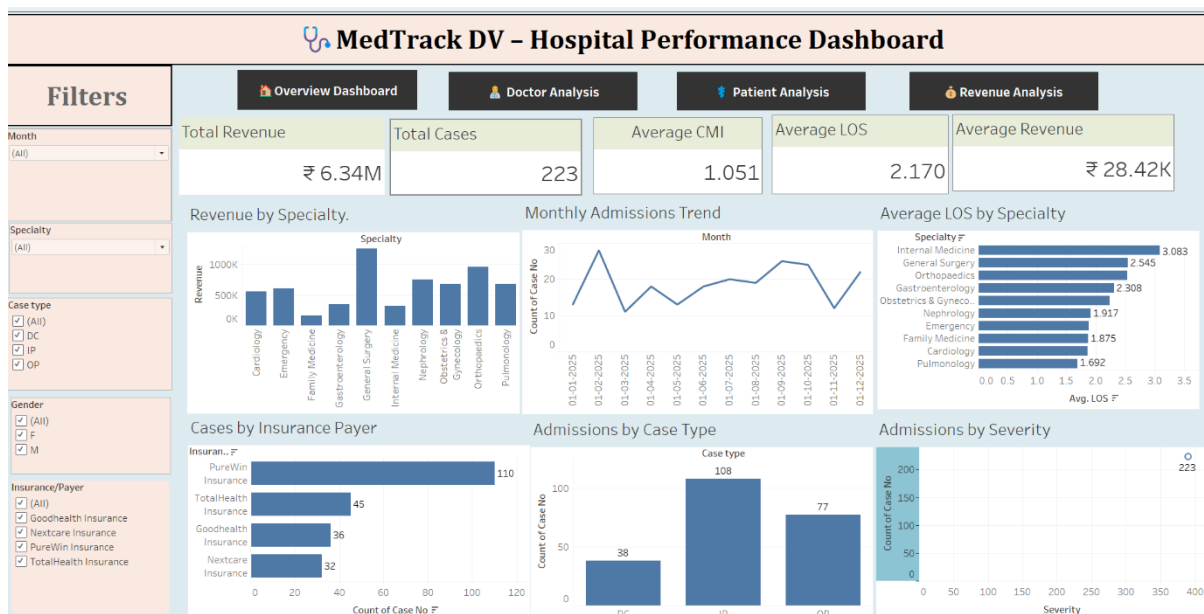
These metrics work together with the dashboard visualizations to provide a broader understanding of both patient operations and financial performance. Interactive filters allow users to examine these metrics for specific months, specialties, case types, genders, doctors, or payer categories. This makes it easier to compare different segments, identify patterns, and understand the factors influencing hospital operations and revenue.

## Dashboard Guide

The **MedTrack DV – Hospital Operations and Patient Analytics** is developed in Tableau and consists of four interactive dashboards: Overview, Doctor Analysis, Patient Analysis, and Revenue Analysis. Each dashboard provides a different perspective of hospital operations and allows users to explore the data using filters and visualizations.

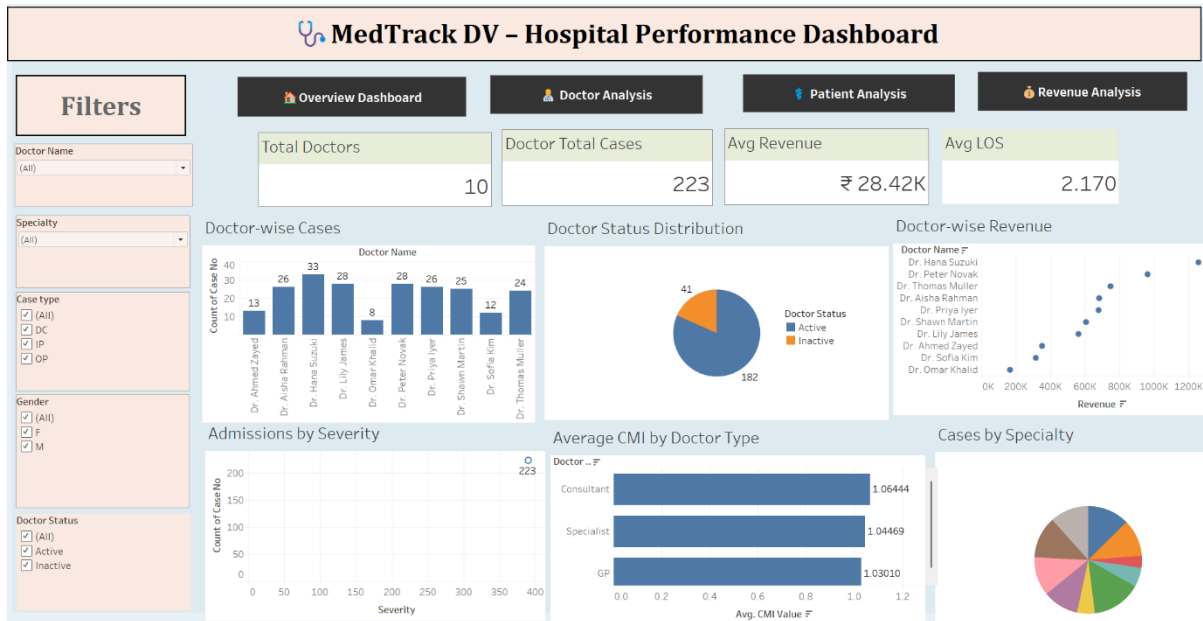
### 1. Overview Dashboard

The Overview Dashboard provides a summarized view of overall hospital performance. It displays important KPIs such as Total Cases, Total Revenue, Average LOS, Average CMI Value, and Total Severity. The dashboard also includes visualizations for monthly admissions, revenue by specialty, severity distribution, gender distribution, case type, average LOS by specialty, and payer distribution. Users can apply filters such as Month, Specialty, Gender, and Case Type to analyze specific segments.



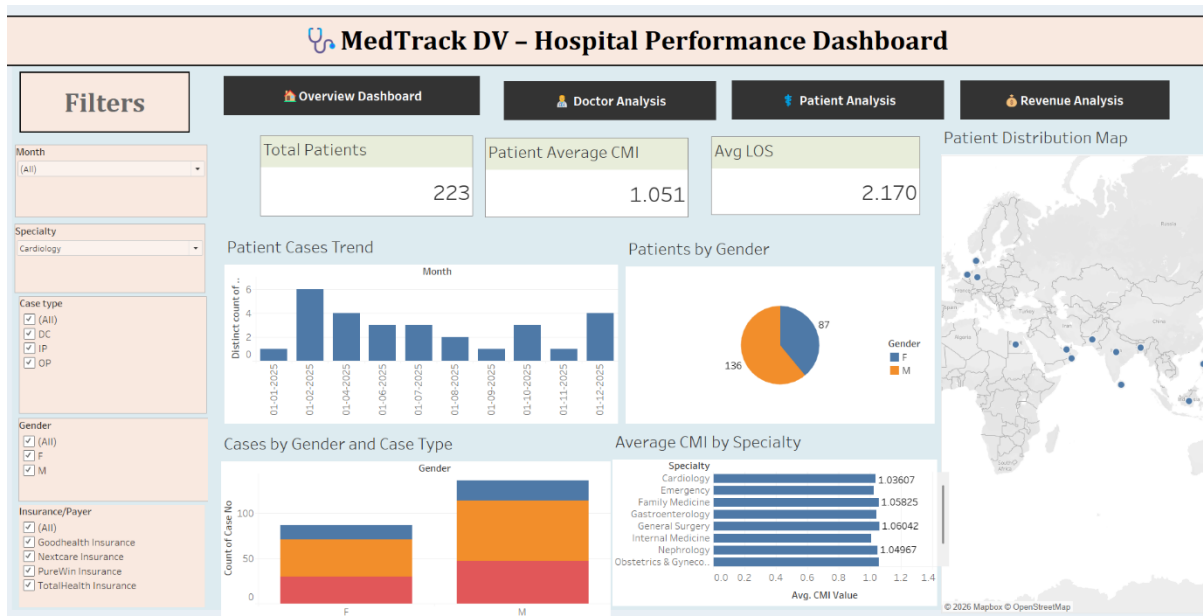
### 2. Doctor Analysis Dashboard

The Doctor Analysis Dashboard focuses on doctor-related operational metrics. It provides insights into doctor case workload, doctor revenue, doctor types, doctor status, CMI values, and specialty distribution. Charts such as doctor-wise cases, revenue analysis, monthly case trends, average CMI by doctor type, and doctor status distribution help users compare doctor-related performance. Filters allow users to select specific doctors, months, specialties, case types, and genders.



### 3. Patient Analysis Dashboard

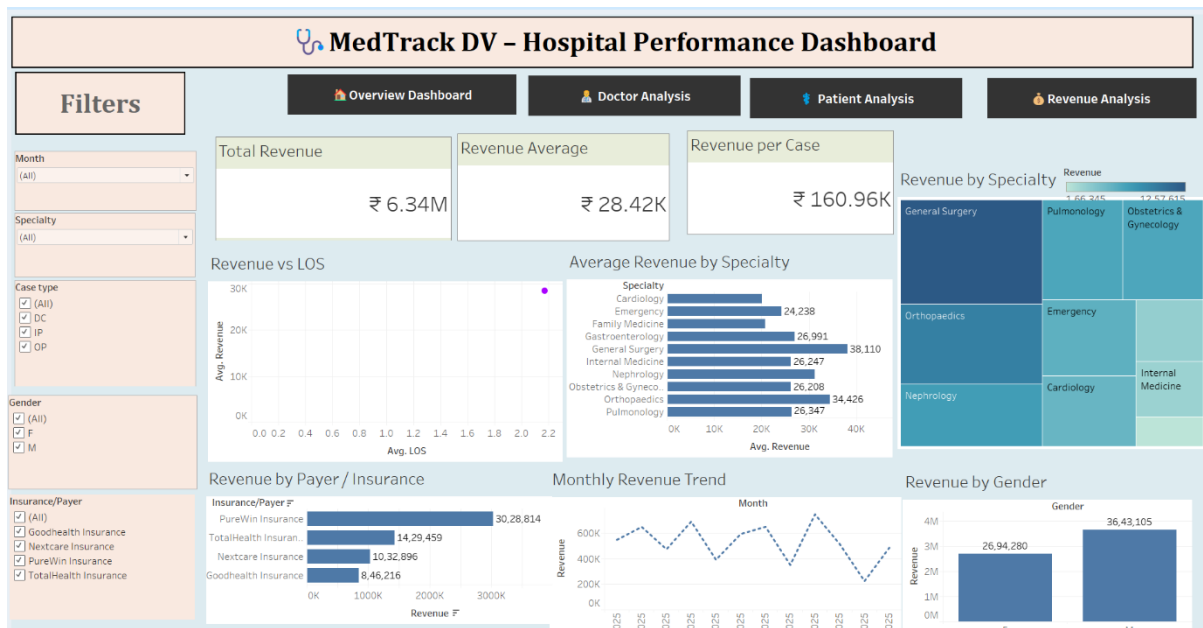
The Patient Analysis Dashboard focuses on patient characteristics and healthcare utilization. It provides metrics such as Average LOS and Average CMI along with visualizations for patient distribution by gender, case type, payer, specialty, and nationality. A geographical map is used to visualize cases by nationality. These visualizations help users understand patient demographics, case distribution, hospital utilization, and case complexity.



### 4. Revenue Analysis Dashboard

The Revenue Analysis Dashboard provides a detailed view of hospital financial performance. It includes KPIs such as Total Revenue and Average Revenue and visualizations for revenue by specialty, case type, payer, and gender. The dashboard also includes a Monthly Revenue Trend and a Revenue vs LOS analysis to explore revenue patterns and relationships. Filters allow

users to analyze revenue for selected months, specialties, case types, and genders.



## Dashboard Navigation & Interactivity

The MedTrack DV – Hospital Performance Dashboard is designed as an interactive Tableau dashboard that allows users to easily navigate between different areas of hospital analysis. The dashboard contains four main sections: Overview, Doctor Analysis, Patient Analysis, and Revenue Analysis. Navigation buttons are provided on the dashboard so users can move from one analysis page to another without having to open individual worksheets separately.

The Overview Dashboard provides a summary of overall hospital performance using important KPIs and visualizations. The Doctor Analysis Dashboard focuses on doctor workload, doctor types, doctor status, case distribution, and CMI-related analysis. The Patient Analysis Dashboard presents information about patient demographics, case types, nationality, payer distribution, length of stay, and case complexity. The Revenue Analysis Dashboard focuses on revenue trends and revenue distribution across specialties, case types, payers, gender, and other categories.

Interactive filters are included to allow users to view specific portions of the dataset. Depending on the dashboard, users can filter the information using dimensions such as Month, Specialty, Gender, Case Type, Doctor Name, and Insurance/Payer. When a filter is applied, the related charts and KPI values are automatically updated to reflect the selected data.

The navigation and filtering features make the dashboard user-friendly, flexible, and easy to explore. Users can move between different analytical views and examine specific data segments without modifying the original dataset. This provides a convenient way to explore hospital operations and identify useful patterns for operational, patient, doctor, and revenue analysis.

## Key Insights

The **MedTrack DV – Hospital Operations and Patient Analytics** provides several insights into hospital operations, patient activity, doctor performance, and revenue patterns. The interactive dashboards allow users to compare different categories and identify areas that may require further attention.

### Overall Hospital Performance

- The dashboard represents **223 hospital cases** in the analyzed dataset.
- The total revenue generated across the available cases is approximately **₹6.34 million**.
- The average **Length of Stay (LOS)** is approximately **2.17 days**, providing an indication of the typical duration of hospitalization.
- The average **CMI Value** is approximately **1.05**, indicating the overall level of case complexity represented in the dataset.
- The total numerical severity value across the records is **390**. This represents the sum of the severity field and should not be interpreted as the number of severe patients.

### Doctor Analysis

- Doctor-wise case analysis helps identify differences in **workload and case distribution** among doctors.
- The dashboard compares **GP, Consultant, and Specialist** doctor categories based on cases and average CMI.
- Doctor status analysis provides a quick view of **active and inactive doctor records**.
- Comparing doctor-related revenue and case volume helps identify variations in workload and financial contribution.

### Patient Analysis

- Patient analysis provides an understanding of case distribution based on **gender, nationality, specialty, case type, and insurance/payer**.
- The nationality map provides a geographical view of the patient population represented in the dataset.
- Analysis of **Length of Stay and CMI** helps understand differences in patient utilization and case complexity across specialties.
- Case-type analysis helps compare different categories such as **IP, OP, and DC** cases.

## Revenue Analysis

- Revenue analysis shows how financial performance varies across **specialties, case types, payers, and gender categories**.
- The monthly revenue trend helps identify changes in revenue over time.
- Revenue by specialty highlights specialties with relatively higher or lower revenue contribution.
- Revenue by payer provides insight into the distribution of revenue across different insurance/payment categories.
- The Revenue vs LOS visualization helps explore whether differences in patient stay duration are associated with differences in revenue.

## Technologies Used

The **MedTrack DV – Hospital Operations and Patient Analytics** was developed using a combination of data preparation, analysis, visualization, documentation, and deployment tools. These technologies helped in transforming the raw hospital dataset into a structured and interactive healthcare analytics solution.

| Technology / Tool           | Purpose                                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Anaconda</b>             | Used as the data science environment for data preparation, cleaning, preprocessing, and exploratory analysis.                       |
| <b>Python</b>               | Used within the Anaconda environment for handling and preprocessing the dataset where required.                                     |
| <b>Pandas</b>               | Used for loading, inspecting, cleaning, and manipulating structured hospital data.                                                  |
| <b>Microsoft Excel</b>      | Used for reviewing, organizing, and maintaining the structured hospital dataset.                                                    |
| <b>CSV / Excel Dataset</b>  | Used as the primary data source containing patient, doctor, clinical, case, insurance, discharge, and revenue information.          |
| <b>Tableau</b>              | Used to create KPI cards, charts, maps, filters, navigation, and interactive dashboards.                                            |
| <b>Tableau Calculations</b> | Used for analytical aggregations such as COUNT, SUM, and AVG to generate KPIs and visualization measures.                           |
| <b>GitHub</b>               | Used to organize project files, documentation, scripts, data, and dashboard-related resources for deployment and portfolio sharing. |
| <b>Tableau Public</b>       | Used/planned for publishing and sharing the completed interactive dashboard online.                                                 |
| <b>Microsoft Word</b>       | Used to prepare the project documentation and supporting project materials.                                                         |

## Deployment

The **MedTrack DV – Hospital Operations and Patient Analytics** project is maintained in a GitHub repository to organize and share the complete project work. The repository contains the different stages of the project, including the raw dataset, data cleaning files, KPI analysis, dashboard storyboard, dashboard images, Tableau workbook, and other project resources.

### GitHub Repository

[Hospital Operations and Patient Analytics Dashboard – GitHub](#)

The project repository is organized into modules based on the development stages:

- **Module-1:** Raw hospital dataset
- **Module-2:** Data cleaning, preprocessing, cleaned dataset, and Jupyter Notebook
- **Module-3:** Final cleaned dataset, KPI analysis, and Power BI file
- **Module-4:** Dashboard storyboard
- **Module-5:** Dashboard design and visualization outputs
- **Module-6:** Tableau workbook and dashboard outputs
- **Module-7:** Final project work and documentation
- **README.md:** General project information

### Tableau Deployment

The final dashboard was developed using **Tableau**. The Tableau workbook is stored in the GitHub repository as a **.twb file**, allowing the dashboard development work to be preserved, reviewed, and shared along with the other project files.

The Tableau workbook is named **Hospital Operations and Patient Analysis Dashboard.twb** and contains the developed dashboard views.

# GitHub Repository Structure

Files

Vanjeri-Akanisha

Go to file

Module-1

hospital\_raw\_data.csv

Module-2

Cleaned dataset.csv

hospital\_cleaning.ipynb

Module-3

Final Cleaned dataset.csv

Hospital generated kpis.ipynb

kpi cards.pbix

Module-4

dashboard\_storyboard.pdf

Module-5

Doctor Analysis.png

Hospital Operations and Patient Analysis Dashboard.twb

Overview Dashboard.png

Module-6

Hospital Operations and Patient Analysis Dashboard.twb

Patient Analysis.png

Revenue Analysis.png

Module-7

Doctor Analysis.png

Hospital Operations and Patient Analysis Dashboard.twb

Overview Dashboard.png

Patient Analysis.png

Revenue Analysis.png

README.md

Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2 / Module-7

alankha28-tech Add files via upload 3725d21 - 1 hour ago History

This branch is 9 commits ahead of and 24 commits behind main.

| Name                                                   | Last commit message  | Last commit date |
|--------------------------------------------------------|----------------------|------------------|
| ..                                                     |                      |                  |
| Doctor Analysis.png                                    | Add files via upload | 1 hour ago       |
| Hospital Operations and Patient Analysis Dashboard.twb | Add files via upload | 1 hour ago       |
| Overview Dashboard.png                                 | Add files via upload | 1 hour ago       |
| Patient Analysis.png                                   | Add files via upload | 1 hour ago       |
| Revenue Analysis.png                                   | Add files via upload | 1 hour ago       |

## Conclusion

The **MedTrack DV – Hospital Operations and Patient Analytics** successfully transforms raw hospital operational data into an interactive, meaningful, and user-friendly analytics solution using Tableau. The primary objective of the project was to analyze hospital data and present important operational, patient, doctor, and financial information through visually appealing dashboards. By applying data cleaning, preparation, analysis, and visualization techniques, the project converts complex hospital data into useful insights that can support better understanding and decision-making.

The developed solution consists of four major dashboards: Overview, Doctor Analysis, Patient Analysis, and Revenue Analysis. The Overview Dashboard provides a consolidated view of hospital performance through important KPIs such as Total Cases, Total Revenue, Average Length of Stay (LOS), Average CMI Value, and Severity. It helps users quickly understand the overall operational condition of the hospital and identify important trends and patterns. The Doctor Analysis Dashboard focuses on doctor-related information, including doctor workload, doctor status, specialties, and case distribution. This helps in understanding workload patterns and can support effective resource allocation.

The Patient Analysis Dashboard provides insights into patient characteristics, case types, severity, specialties, gender, nationality, and other available patient-related information. These visualizations help users identify common patient patterns and understand the distribution of hospital cases. The Revenue Analysis Dashboard focuses on financial information such as revenue distribution, insurance and payer details, payer mix, and case-related revenue patterns. This enables users to understand the major contributors to hospital revenue and supports financial and operational analysis.

Overall, the project demonstrates the effective use of Tableau for healthcare data visualization and analysis. It provides a centralized platform for monitoring hospital performance, analyzing doctor and patient information, understanding revenue patterns, and supporting operational planning. The dashboard reduces the complexity of raw data by presenting information through KPIs, charts, graphs, filters, and interactive dashboard components. It can therefore assist hospital administrators and decision-makers in identifying trends, monitoring performance, planning resources, and making data-driven decisions.